STA 138 Project

An Analysis of Byssinosis Development

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Background

Byssinosis is a lung disease, but more specifically, it is a form of pneumoconiosis that is at elevated risk among textile workers. Textile workers have experienced a number of factors that may contribute to their risk of contracting lung disease. The question we are interested in answering is which of these factors are significant in developing byssinosis. In order to explore this, we will consider a variety of generalized linear models using logistic regression to explore what variables are significant in the development of byssinosis. In other words, we are interested in understanding what risk factors exist and contribute the most to contracting byssinosis and what model best explains and provides clarity on the development of byssinosis within these textile workers.

Initial Data Analysis

The data that we will use to develop our model and explore risk factors is a 1973 dataset from a cotton textile company based in North Carolina. The company was interested in measuring the prevalence of byssinosis among its workers (n = 5,418) as well as recording the following variables: years of employment (less than 10, 10-19, and 20 or more), smoking (in the last 5 years or not), dustiness of the workplace (1 = most dusty, 2 = less dusty, 3 = least dusty), race (white or other), and sex (male or female). Our dataset is fairly representative of each level of the variables; sex has a 53/47 split, smoking has a 58/42 split, workplace dustiness has a 12/24/64 split, race has a 65/35 split, and years of employment has a 50/13/37 split. Since the sample size is so big, we do not need to worry significantly about the underrepresentation of certain variables. Out of the entire log of workers, about 3% of them contracted byssinosis.

By looking at the dataset, we see that the highest likelihood of developing byssinosis comes from white men who smoke, work in a dusty workplace, and have worked for over 20 years in textile factories. However, this is a very broad range of variables, so we will need to isolate the effects of each variable to understand which are risk factors for byssinosis. Exploring the differences between other combinations of variables indicates that the dustiness of the workspace, smoking status, and years of working might be risk factors for the development of byssinosis. However, we would like to explore further statistical testing to ensure that these variables are significant in developing a model for byssinosis.

Training and Testing Models

Although stepwise regression can be flawed, it is often an effective method for generating a logistic regression model. Since we do not have the computing to perform all subset selection, we must rely on stepwise regression to find what appears to be the most explanatory model. For the sake of reducing bias, we will be training and testing our data on itself to understand whether our model is overfitting or unrepresentative of the data. For our testing and training sets, we decided to use an 80:20 split. We did this because the data has a large sample size of 5418 and allows our logistic regression model to understand patterns and relationships while also testing them on a sufficiently large amount of data. Since there are a number of factors in our data, we prefer to use AIC in our stepwise regression since it will be important to understand if risk factors exist in our data, even if they are less risky than other variables. After training our model we obtained this logistic regression model:

$$\text{logit}(\pi_i) = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \beta_3 x_{i3} + \beta_4 x_{i4} + \beta_5 x_{i5}$$

Where π_i denotes the probability of contracting byssinosis, x_{i1} is an indicator if the textile factory has a ranking of 2 in workplace dustiness or not (1 or 0), x_{i2} is an indicator if the textile factory has a ranking of 3 in workplace dustiness or not (1 or 0) x_{i3} is an indicator if the employee has worked there for 20+ years or not (1 or 0), x_{i4} is an indicator if the employee has worked there for 10-19 years or not (1 or 0), and x_{i5} is an indicator if the person has smoked in the last 5 years (1 or 0).

After developing our model, we must fit the logistic regression on the remaining 20% of our data to determine our parameter estimates. By doing this we obtain the following estimates:

$$\beta_0 = -1.70189,\ \beta_1 = -2.20451,\ \beta_2 = -3.07987,\ \beta_3 = .45892,\ \beta_4 = -.24016,\ \beta_5 = .01827$$

We can immediately recognize that some of our parameter estimates are more significant than others, as β_5 is substantially closer to 0 than β_2 (their standard errors are approximately .4). Alongside generating our estimates, we also received Wald test p-values indicating whether or not they were significant. Even though β_3 through β_4 are nonsignificant, we are interested in keeping these parameters in our model since they substantially affect our log odds. Furthermore, these estimates are related to the amount of time spent in textile factories, so it is worth exploring their significance beyond Wald testing. In order to do this, we perform a likelihood ratio test where our reduced model is the suggested model under what our Wald test p-values suggest, while our full model is the model using all the estimates above.

Model Selection

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Analysis of Deviance Table

Model 1: Byssinosis ~ Workspace + Employment
Model 2: Byssinosis ~ Workspace + Employment + Smoking
  Resid. Df Resid. Dev Df  Deviance Pr(>Chi)
1      1079     246.10
2      1078     246.09  1 0.0023397   0.9614
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Analysis of Deviance Table

Model 1: Byssinosis ~ Workspace2 + Workspace3 + +Employment_20
Model 2: Byssinosis ~ Workspace + Employment + Smoking
  Resid. Df Resid. Dev Df Deviance Pr(>Chi)
1      1080     246.26
2      1078     246.09  2  0.16947   0.9188
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In order to figure out the significance of these estimates, we conducted multiple likelihood ratio tests with the various coefficients. Our first test was conducted by comparing the full model that included the predictors of Workspace, Employment, and Smoking, and compared it to a reduced model of just Workspace and Employment. The only difference between the two models was the additional predictor variable of Smoking, and as we can see from our initial results, there is only a marginal difference in deviance. Therefore, the high p-value indicates adding smoking to our model does not enhance the explanation of byssinosis development.

From this first model, we wanted to examine our estimates more closely, especially by comparing them to the full model which includes Workspace, Employment, and Smoking. We then continued to look into the significant coefficients like Workplace 2, Workplace 3, and Employment_20; Workplace takes into account the different levels of dustiness, while Employment_20 represents if that particular person has worked at their job for more than 20 years. We decided to include their predictors as they made the most sense in what could contribute to Byssinosis and showed up as the biggest contributors in the initial logistic regression model. We chose to remove some of the other coefficients due to the slope of the beta values and how it was close to 0 which meant they had little to no significance in our model. After comparing the full model to those 3 predictors, the ANOVA table, including a summary of our model, tells us that the Workplace and Employment_20 estimates are indeed highly significant to our model.

Conclusion

Our final model of byssinosis development was an additive logistic regression model dependent on 3 covariates: x_{i1}(whether or not the workplace is given a 2 for dustiness level), x_{i2} (whether or not the workplace is given a 3 for dustiness level), and x_{i3} (whether or not the employee has worked at the textile factory for over 20 years). In order to develop this model, we used an 80:20 training and testing split and performed stepwise regression on our training set. From there, we fit the model to our testing set and obtained some parameter estimates, although not all of them performed significantly in Wald tests. This is likely because the goal of stepwise regression is to get the smallest AIC possible, rather than developing significant parameters. Furthermore, our 80:20 split resulted in a lack of power in these Wald tests which may explain the insignificant p-values. This motivated us to explore likelihood ratio tests to attempt to retain the insignificant parameters. We did this by analyzing if the deviance is significantly different between the reduced and full models. Finally, we obtained this model:

$$\mathrm{logit}(\pi_i) = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \beta_3 x_{i3}$$

with parameter estimates:

$$\beta_0 =- 1.70189,\ \beta_1 =- 2.20451,\ \beta_2 =- 3.07987,\ \beta_3 = .45892$$

This model adequately represents the risk factors associated with byssinosis development, those being: dustiness of the workplace and years working in the textile industry. By being in the least dusty workplace, your chance of contracting byssinosis decreases by an approximate log odds of 3.08. After working in a textile factory for 20+ years, your chance of contracting byssinosis will increase by an approximate log odds of .459. To conclude, workplace dustiness and years employed in a textile factory significantly affect your chances of developing byssinosis, with our model approximating a 22% chance if both risk factors are present.